PRINTFORGE INDUSTRIAL PRESS FACILITY

TECHNICAL PREFLIGHT & PRESS-READY FILE SPECIFICATIONS - REVISION 2026.4

1. Mandatory PDF/X Standards & Color Spaces

All production press files must be submitted conforming to PDF/X-4:2010 or PDF/X-1a:2001 standards.
- Color Mode: strictly CMYK and validated Pantone Matching System [PMS] Spot Colors.
- RGB, LAB, and DeviceN color profiles will be auto-converted to Fogra 51 / GRACoL 2013.
- Total Area Coverage [TAC / TIC]: Maximum 320% on Coated stock, 280% on Uncoated text.
- Black Generation: Use Rich Black [C:60% M:40% Y:40% K:100%] for large solids; 100% K for body text.

PARAMETER	MINIMUM SPECIFICATION	TOLERANCE / CRITICAL THRESHOLD
Bleed Margin	3.175 mm [0.125 in] on all 4 sides	5.0 mm required for Hardcover cases
Safety Safe Zone	4.5 mm [0.177 in] inside trim edge	Keep text & logos clear of fold/score lines
Image Resolution	300 DPI at 100% final output size	Bitmaps & Line Art must be 1200 DPI
Vector Line Weight	0.25 pt minimum [0.088 mm]	0.50 pt minimum for Reverse/Knockout text
Font Embedding	100% Embedded or Outlined/Curved	TrueType/Type 1 must be OpenType/OTF

2. Spot Finishes, Die-Lines & Embellishments Setup

To ensure automated optical die-cutting and Kurz foil registration, follow layer naming standards:
1. Hot Foil Stamping: Create Spot Color named [Foil-Gold] or [Foil-Silver], set to 100% Magenta, OVERPRINT ON.
2. Raised Spot UV: Create Spot Color named [Spot-UV], 100% Cyan, set OVERPRINT ON on dedicated vector layer.
3. Structural Die-Cut: Create 0.5 pt Spot Color line named [Die-Cut] [100% Magenta], 100% stroke.
4. Mechanical Crease: Create 0.5 pt Spot Color line named [Crease] [100% Cyan], dashed 4pt stroke.
5. Kiss-Cut / Sticker: Create Spot Color named [Kiss-Cut] [100% Green], 0.25 pt stroke.

3. Spine Bulking & Signature Formula Matrix

Calculate spine thickness prior to export using our calibrated Heidelberg press bulking factors:
Spine Width = [Page Count / 2] x Paper Caliper [mm] + Board Caliper [for Hardcover]
- 80 GSM Munken Print Cream: Caliper = 0.120 mm [Bulk factor 1.50]
- 115 GSM Silk Coated Art: Caliper = 0.098 mm [Bulk factor 0.85]
- 150 GSM Heavy Matte Art: Caliper = 0.138 mm [Bulk factor 0.92]
- 3.0 mm European Dense Kappa Binder Board adds 6.0 mm overall to case spine calculations.

PREPRESS ASSISTANCE & DIRECT INPRESS LINE

Have questions about creep calculation, gatefold dielines, or Delta E tolerances? Contact our prepress lab:
Direct Press Hotline: +1 [800] 555-7746 | Prepress FTP: prepress@printforge-demo.com

PRINTFORGE PREFLIGHT VERIFICATION CHECKLIST

14-POINT AUTOMATED PRODUCTION QUALITY ASSURANCE - ISO 12647-2 CERTIFIED

Stage 1: Document Architecture & Geometry Checklist

[PASS] 01. Trim Box, Bleed Box, and Media Box accurately defined in PDF header attributes.
[PASS] 02. Minimum 3.175 mm bleed provided on all exterior trim edges without truncation.
[PASS] 03. Safe margins of 4.5 mm strictly preserved from all page perimeter cutting lines.
[PASS] 04. Multi-page booklet files submitted as single consecutive pages [NOT printer spreads].
[PASS] 05. Saddle-stitched publications pagination is a multiple of 4 [4, 8, 12, 16, 20...].
[PASS] 06. Perfect bound & Smyth-sewn signatures paginated in 8, 16, or 32-page sections.

Stage 2: Color, Inking & Screening Verification

[PASS] 07. All photographic assets converted to ISO Coated v2 [FOGRA39/51] with Delta E < 1.5.
[PASS] 08. Zero RGB, LAB, or untagged device color spaces present in vector or raster assets.
[PASS] 09. Total Ink Limit [TIL] does not exceed 320% on Coated and 280% on Uncoated sheets.
[PASS] 10. Small black body text [< 18 pt] is 100% Black [C:0 M:0 Y:0 K:100] with Overprint ON.
[PASS] 11. White reverse text on dark backgrounds has Overprint OFF to prevent knockout bleed.

Stage 3: Typography, Resolution & Embellishments

[PASS] 12. All fonts 100% embedded as subsets or converted into vector line curves.
[PASS] 13. Raster artwork resolution verified >= 300 DPI at 100% final output reproduction.
[PASS] 14. Spot finishes [UV, Foil, Emboss, Dieline] configured on dedicated layers with OVERPRINT.

HEIDELBERG PRINECT AUTOMATED WORKFLOW CERTIFICATION

This job specification complies with ISO 12647-2 commercial offset standards and is certified for automated CTP plate imaging on Kodak Sonora 2400 DPI thermal systems.

Inspection Engineer: Dr. Evelyn Sterling, Head of Color Science & Prepress
Plant Facility: PrintForge Press Plant 1, 742 Industrial Pkwy, New York, NY
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